Reg. No.															

M. Tech. (PT) DEGREE EXAMINATION, JANUARY 2024

Second Semester

20PITE53J – TIME SERIES FORECASTING

(For the candidates admitted during the academic year 2020 – 2021 onwards)

Time: Three Hours | Max. Marks: 100

Answer **ALL** Questions

PART – A (10 × 2 = 20 Marks)

		Marks	BL	CO	PO
1.	What is Seasonality?	2	1	1	1
2.	List the different components of time series?	2	1	1	1
3.	Explain Naive forecast method?	2	1	2	1
4.	List some limitations of the Moving Average Forecast method?	2	1	2	1
5.	What is the Autocorrelation function?	2	1	3	2
6.	List some properties of stationary time series?	2	1	3	2
7.	What are the components of the ARIMA model?	2	1	4	2
8.	What is the SARIMA model?	2	1	4	2
9.	What is the VARMA model?	2	1	5	2
10.	Define Multivariate Time series Analysis?	2	1	5	2

PART – B (5 × 16 = 80 Marks)

		Marks	BL	CO	PO
11.a.i.	What is Forecasting?	2	1	1	1
ii.	Why do we need forecasting?	2	1	1	1

iii.	In how many types can Forecasting techniques be classified? Explain any one type in detail.	4	1	1	1
iv.	Explain components of time series in detail?	8	1	1	1
	(OR)				
b.i.	Discuss the need of decomposition in time series? Explain the decomposition models in detail.	8	1	1	1
ii.	What is not a time series? Give some applications of time series data? Explain the features of time series data?	8	1	1	1
12.a.i.	Explain the simple forecast technique with its types in detail?	8	1	2	1
ii.	What is moving average? What are its limitations?	8	1	2	1
	(OR)				
b.i.	What are the Measures of forecast accuracy? Explain with respective formulae.	3	3	2	3
ii.	Discuss some points about splitting time series data.	3	3	2	3
iii.	Explain Holt-winter Model?	5	3	2	3
iv.	Explain Holt- Model?	5	3	2	3
13.a.i.	Explain Autocorrelation function?	3	2	3	1
ii.	Explain Partial Autocorrelation function for ARMA?	3	2	3	1

iii. Explain Dickey-Fuller test.	3	2	3	1
iv. Explain box-cox transformation. How do you make data stationary with box-cox transformation?	7	2	3	1
(OR)				
b. Explain order "pq" time series process and its construction in detail. Also explain order "p" and order "q"	16	1	3	2
14.a. What is ARIMA differencing? Explain its types in detail.	16	1	4	2
(OR)				
b. Explain SARIMA model and its parameter combinations in detail? Describe the model construction with suitable example.	16	1	4	2
15.a.i. What are the different statistical approaches in time series analysis? Explain with suitable examples?	10	1	5	2
ii. Explain (i) Vector Time Series model (ii) Need of cointegration	6	1	5	2
(OR)				
b. Describe the VARMA model? Explain its need and construction in detail with suitable examples?	16	1	5	2

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